Gengcong Yang

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Education

University of Texas at Austin (UT Austin) — 08/2023 – 12/2024
*Master in **Computer Science**; GPA: 3.7 / 4.0* — Austin, United States

Tsinghua University (THU) — 08/2019 – 06/2022
*Master in **Control Engineering**, Department of Automation; GPA: 3.6 / 4.0* — Beijing, China

National Chiao Tung University (NCTU) — 09/2017 – 01/2018
*Exchange Student in **Computer Science**; GPA: 3.8 / 4.0* — Hsinchu, Taiwan

Sun Yat-sen University (SYSU) — 08/2015 – 06/2019
*Bachelor in **Software Engineering**; GPA: 3.8 / 4.0* — Guangzhou, China

Publications

Diamond: Harnessing GPU Resources for Scientific Deep Learning — 2025

- Authors: Haotian Xie, Rohan Marwaha, Minu Mathew, Song Bian, **Gengcong Yang**, et al.
- Published in eScience

Probabilistic Modeling of Semantic Ambiguity for Scene Graph Generation — 2021

- Authors: **Gengcong Yang**, Jingyi Zhang, Yong Zhang, Baoyuan Wu, Yujiu Yang
- Published in CVPR, **one of the top conferences in artificial intelligence**

Experiences

TikTok USDS, Emerging Product, *Machine Learning Engineer* — San Jose, 07/2026 – Present

- Coordinating model evaluation for Seedance, a video generation model, working cross-functionally to validate model performance ahead of U.S. rollout.
- Supporting model serving infrastructure for Seedance's U.S. deployment, partnering with internal MaaS and external cloud providers on provisioning, monitoring, and SLA compliance.

Uber, Driver Offer Pricing Team, *Machine Learning Engineer* — Sunnyvale, 01/2025 - 07/2026

- Led optimization of ML observability and reliability infrastructure for pricing prediction pipelines, improving production stability and compute efficiency at scale.
- Developing driver fare prediction models that serve as core pricing signals for rider-side pricing systems, directly influencing marketplace efficiency.
- Enhanced reliability and efficiency of ML-driven pricing systems supporting millions of trips and real-time economic decision-making at national and global scale.

Texas Advanced Computing Center, *Graduate Research Assistant* — Austin, 01/2024 – 12/2024

- Worked with a team on the early-stage development of Diamond, a web platform to support the entire Machine Learning lifecycle.
- Developed a job submission pipeline enabling users to build images and train ML models across multiple nodes.

Electronic Arts, *Software Engineer Intern* — Austin, 05/2024 – 08/2024

- Collaborated with the Madden Systems team to enhance system efficiency and resolve bugs.
- Centralized PS5 Activities configuration, eliminating redundancy and reducing debugging time.

ByteDance, E-commerce, *Machine Learning Engineer* — Shanghai, 06/2022 – 06/2023

- Developed image search models and collaborated on a Retrieval-Ranking framework for the TikTok Shop similar product search project, contributing to a 30%+ increase in key evaluation metrics.
- Constructed datasets of similar product images for Post-Pretraining, containing over 10 million classes.
- Led labeling tasks for similar image pairs and fine-tuned models, resulting in a 10%+ boost in model performance.
- Quantized object detection models using QAT and reduced latency by 75% while preserving accuracy.

Tencent, *Machine Learning Engineering Intern* — Beijing, 06/2021 – 08/2021

- Focused on the retrieval stage of the recommendation system for an early-stage short-video platform.
- Built an offline evaluation pipeline, enabling efficient initial assessment of retrieval models against historical data.
- Experimented with Pairwise training for the two-tower model, achieving a 5%+ improvement in offline metrics.

KAUST, *Research Intern* — Remote, 02/2021 – 07/2021

- Implemented a differentiable RoI Pooling method with CUDA and C++ for temporal action detection task on videos and achieved experimental improvement by 6%.

Tencent AI Lab, *Computer Vision Researcher Intern* — Shenzhen, 07/2019 – 07/2020

- Implemented a novel modeling method for polysemous representation and applied for a technology patent.
- Designed a probabilistic algorithm for the uncertainty modeling of visual scenes and published a paper in CVPR.

Patents

Visual Relationship Recognition Method Based on Artificial Intelligence

- Inventors: **Gengcong Yang**, Yong Zhang, Baoyuan Wu, Yanbo Fan, Zhifeng Li, Yujiu Yang, Wei Liu
- Patent Number: CN202011108484
- Granted Date: 03/05/2024

Technical Skills

Programming Languages: Python, C++, SQL, Java, JavaScript

Tools & Technologies: TensorFlow, PyTorch, Recommendation Systems, NLP, CV, CUDA, Hadoop, Spark, Docker, Git, AWS

Selected Awards

SYSU Excellent Scholarship Third Prize	2016, 2017
Interdisciplinary Contest In Modeling Honorable Mention	2017
SYSU Xian Weijian Overseas Study Funding	2017
National Encouragement Scholarship	2017, 2018
SYSU Excellent Scholarship Second Prize	2018
SYSU Outstanding Undergraduate Thesis Award	2019
Tsinghua Excellent Scholarship First Prize	2021
Tsinghua SIGS Professional Practice Scholarship	2022